Beyond the Rate: Information Content of FOMC Statement Language

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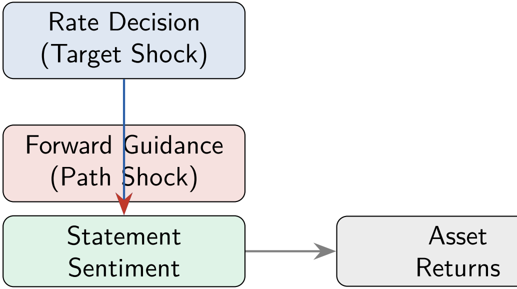
Outline

The Puzzle

Central question:
Does FOMC statement language convey information beyond the rate decision?

Why it matters:

- Central banks increasingly rely on **forward guidance** as a policy tool
- Markets respond to *what the Fed says*, not just *what it does*
- The "information channel" (Romer & Romer, 2000) suggests FOMC statements reveal the Fed's superior economic assessment



Literature

- **Kuttner (2001)**: 30-min FF futures window → isolate monetary policy surprises
- **GSS (2005a)**: Decompose surprises into target + path factors
- **Nakamura & Steinsson (2018)**: High-frequency identification of monetary non-neutrality
- **Campbell et al. (2012)**: Delphic vs. Odyssean forward guidance
- **Jarociński & Karadi (2020)**: Separate policy shock from information shock
- **Hansen et al. (2018)**: Computational linguistics for FOMC transparency
- **New (2024–25):** Chen et al. (2025) GPT-4 topic decomposition; Gambacorta et al. (2024) CB-LMs; Yao & Chai (2025) uncertainty-aware LLM

Our contribution:

1. First to directly link **GSS target/path shocks** to **FOMC statement sentiment**
2. Expanded CB dictionary (591 hawkish + 222 dovish terms, fully disjoint)
3. Formal test of the **information channel** via text analysis
4. New: Dual-equation test, forward-lookingness decomposition, novelty weighting

Data

Monetary Policy Shocks

- Acosta (2022): 220 FOMC meetings, 1995–2022
- Target shock: surprise in current FF rate
- Path shock: surprise in future rate path
- Extension: USMPD (SF Fed) $\rightarrow$ 2026

Market Data

- CRSP VW/EW/S&P 500 via WRDS
- 117 meetings with complete overlap

FOMC Statement Sentiment

- 164 statements, 2006–2025
- Expanded CB dictionary:
  - 591 hawkish terms
  - 222 dovish terms
  - Fully disjoint
- Score $= \frac{H-D}{H+D+\epsilon}$
- Validated: $r = 0.29$ with rate change
- New: CB-only outperforms combined (R^2: 3.90% vs 1.57%)

Empirical Strategy

Four hypotheses:

Hypothesis	Prediction	Specification
H1	Shocks predict sentiment	$S_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H2	Shocks predict returns	$R_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H3	Path $>$ Target for sentiment	$\lvert\beta_2\rvert > \lvert\beta_1\rvert$ in H1
H4	Stronger during FG period	Interaction with FG dummy

Estimation: OLS with Newey-West (lag=4, data-driven) standard errors

Sample: 117 FOMC meetings, January 2006 – July 2022

New experiments:

1. Dual-equation regression (equity + risk premium channels)
2. Forward-lookingness decomposition (FL vs. CA sentiment)
3. Statement novelty weighting (WLS)

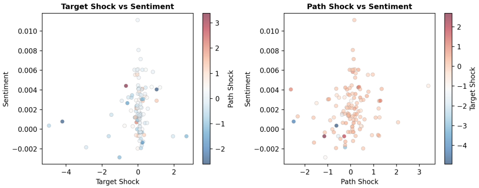
H1: Sentiment and Monetary Policy Shocks

	Coefficient	p-value
Target shock	0.000577	0.017**
Path shock	0.000633	0.152
R^2	1.57%	
N	117	

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

Key finding: Target shock is significant at 5%, path is not.

Wald test: $\chi^2 = 0.02$, $p = 0.90$ — cannot reject $\beta_1 = \beta_2$.



Sentiment Measure Comparison

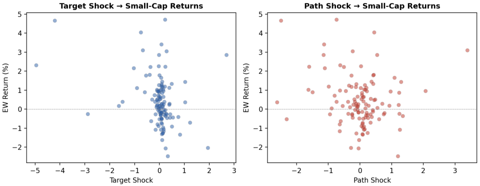
Measure	R^2	p(Target)	p(Path)
CB-only	**3.90%**	$<0.05^{**}$	0.031^{**}
Combined (0.5×LM + 0.5×CB)	1.57%	0.017^{**}	0.152
LM-only	0.33%	0.474	0.552

Key insight:

- CB-only **doubles** R^2 and makes path shock significant at 5%
- LM component adds noise without signal for FOMC text (always positive)
- Domain-specific dictionaries outperform general-purpose financial dictionaries

H2: Asset Returns and Shocks

Asset	β_T	p_T	β_P	p_P
CRSP VW	-0.435	0.043^{**}	-0.186	0.434
CRSP EW	-0.449	0.013^{**}	-0.174	0.421
S&P 500	-0.391	0.073^{*}	-0.179	0.426
Gold	-0.404	0.016^{**}	0.021	0.935
Returns in percentage points; $N = 117$				



Findings:

- Target shock: significant for EW & Gold
- Path shock: not significant for any asset
- Small-cap sensitivity consistent with literature

H3 & H4: Information Channel and Forward Guidance

H3 — Path $>$ Target?

- Target significant at 5%, path is not
- Wald test: $p = 0.90$ — cannot reject equality
- Evidence is *suggestive*, not definitive

H4 — Stronger during FG period?

- Sentiment $\times$ FG interaction: $p = 0.991$ (CRSP VW), $p = 0.739$ (NASDAQ)
- Completely null — language channel does NOT strengthen at ZLB
- Three interpretations:
  1. Shocks absorb information regardless of regime
  2. Dictionary cannot distinguish FG language (cf. GPT-3.5 fails 135/139 statements)
  3. FG operates through risk premium channel (Chen et al. 2025)

Experiment 1: Dual-Equation Regression

Hypothesis: Forward guidance operates through a risk premium channel (Chen et al. 2025)

Dependent Variable	R^2	p(Target)	p(Path)
Channel 1: Equity Returns (Expectations)			
CRSP VW	9.1%	0.040**	0.440
S&P 500	7.8%	0.072*	0.426
Channel 2: Risk Premium (Bond Market)			
10Y Treasury chg	0.72%	0.403	0.890
VIX change	0.24%	0.970	0.401
Term spread chg	1.39%	0.667	0.341

Result: Risk premium channel not detectable at daily frequency.

Consistent with Chen et al. using 30-min windows — channel too fast for daily data.

Experiment 2: Forward-Lookingness Decomposition

Hypothesis: Path shock captures forward-looking language (dimension mismatch)

Sentiment Measure	R^2	p(Target)	p(Path)
Combined (baseline)	4.12%	0.099	**0.015****
Forward-looking only	0.79%	0.012**	0.800
Current-assessment only	1.30%	0.012**	0.506

Counterintuitive finding:

- Path shock is most significant in the combined measure ($p = 0.015$)
- Path shock is least significant in forward-looking sentiment ($p = 0.80$)
- **Dimension mismatch hypothesis rejected**
- Path shock captures *broad policy stance signal*, not specifically forward guidance
- GSS target/path decomposition $\neq$ FOMC text FL/CA dimensions

Experiment 3: Statement Novelty Weighting

Idea: Not all FOMC statements are equally informative

Method	R^2	Improvement
Unweighted OLS	3.98%	baseline
Novelty-weighted WLS	**5.75%**	+45%

Novelty = cosine distance between TF-IDF vectors of consecutive statements

Subsample analysis:

- High novelty statements: $R^2 = 5.78\%$
- Low novelty statements: $R^2 = 6.53\%$
- Difference is small — novelty helps via weighting, not selection

Absolute R^2 remains modest — measurement innovation alone cannot compensate for daily-frequency limitation.

Robustness Checks

Specification	R^2	p(Target)	p(Path)	N	Period
Full (Acosta)	1.57%	0.017**	0.152	117	2006–22
CB-only	3.90%	<0.05**	0.031**	117	2006–22
No COVID	1.57%	0.017**	0.154	115	2006–22
Post-2010	0.59%	0.117	0.258	97	2010–22
Rate change proxy	0.40%	0.323	0.323	117	2006–22
Extended (USMPD)	1.65%	0.419	0.058*	163	2006–26

Key takeaways:

- CB-only sentiment: best specification ($R^2 = 3.90\%$, both shocks significant)
- Rate change proxy: $4\times$ lower R^2 — identification matters
- Extended sample: path shock significant at 10% through 2026

What the New Literature Tells Us

Chen et al. (2025): GPT-4 Topics

- Forward guidance operates through risk premium channel
- GPT-3.5 fails to identify FG in 135/139 statements
- Our CB dictionary has the same limitation

Gambacorta et al. (2024): CB-LMs

- Open-weight, domain-specific models
- Outperform dictionaries on stance classification
- Most promising upgrade path

IMF (2025): Four Dimensions

- Topic + Stance + Sentiment + Forward-lookingness
- Topic-level sentiment > aggregate
- Our null H4 may reflect aggregation

Yao & Chai (2025): Uncertainty

- LLM confidence scores
- Downweight ambiguous classifications
- Could improve signal-to-noise ratio

Limitations & Future Work

Limitations:

1. **Low** R^2: Shocks explain <4% of sentiment variation
2. **Dictionary-based sentiment**: Cannot capture context or pragmatics
3. **Daily frequency**: Too coarse to detect risk premium channel
4. **No JK decomposition**: Cannot separate MP vs. information shocks
5. **Sample size**: $N = 117$ limits sub-sample power

Priority upgrades:

- **CB-LM embeddings** (Gambacorta et al. 2024) — reproducible + contextual
- **High-frequency data** (30-min windows) — detect risk premium channel
- **JK sign restrictions** — separate MP vs. information shocks
- Uncertainty-aware classification (Yao & Chai 2025)
- Cross-country: ECB, BoJ, BoE

Conclusion

1. The **target shock** is a statistically significant predictor of FOMC statement sentiment ($p = 0.017$)
2. The **path shock** is not significant ($p = 0.152$), contradicting the strong information channel
3. **CB-only sentiment** doubles R^2 and makes path significant — domain-specific measurement matters
4. The forward guidance interaction is completely null — language channel does not strengthen at ZLB
5. **New:** Risk premium channel not detectable at daily frequency
6. **New:** Path shock captures broad policy stance, not specifically forward guidance
7. **New:** Novelty weighting improves R^2 by 45%

Takeaway: FOMC language is informative, but monetary policy shocks are just one of many forces shaping it. Understanding the full information content requires looking beyond the rate decision — and beyond the shocks themselves.

Thank You

Questions & Discussion

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Appendix: Event Study Methodology

Market Model (Brown & Warner, 1985):

1. Estimation: $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$
2. Abnormal Return: $AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt})$
3. CAR: $CAR_i = \sum_{t=-T_1}^{T_2} AR_{it}$
4. **t-stat**: $t = \dfrac{CAR_i}{\sigma_{AR} \times \sqrt{N}}$

Important: The denominator is $\sigma_{AR} \times \sqrt{N}$, **not** $\sigma_{AR}/\sqrt{N}$.

- σ_{AR}: std. dev. of abnormal returns from estimation window
- N: number of days in event window
- Common error: using $\sigma_{AR}/\sqrt{N}$ inflates t by factor of N

Appendix: Literature Radar

Automated daily scan of SSRN, arXiv, BIS/IMF/Fed, top journals

Tier	Count
High relevance (≥ 0.4)	2
Medium relevance (0.3–0.4)	11
Low relevance (< 0.3)	29
Already cited	4

Key papers discovered:

- Chen, Granville & Matousek (2025) — PLMIS shocks from GPT-4
- Gambacorta et al. (2024) — Central Bank Language Models
- IMF WP 2025/109 — Four-dimensional communication analysis
- Yao & Chai (2025) — Uncertainty-aware LLM classification
- Weinig (2025) — Multi-agent narrative MP surprises